detected experimentally in 1956 by Fredrick Reines and Clyde Cowan. Later Reines received Nobel prize in physics in the year 1995 for his discovery.

The neutrino has the following properties

- It has zero charge
- It has an antiparticle called anti-neutrino.
- Recent experiments showed that the neutrino has very small mass.
- It interacts very weakly with the matter. Therefore, it is very difficult to detect it. In fact, in every second, trillions of neutrinos coming from the sun are passing through our body without causing interaction.

9.6.3 Gamma emission

In α and β decay, the daughter nucleus is in the excited state most of the time. The typical life time of excited state is approximately 10^{-11}s. So this excited state nucleus immediately returns to the ground state or lower energy state by emitting highly energetic photons called γ rays. In fact, when the atom is in the excited state, it returns to the ground state by emitting photons of energy in the order of few eV. But when the excited state nucleus returns to its ground state, it emits a highly energetic photon (γ rays) of energy in the order of MeV. The gamma emission is given by

$$ {}^{A}_{Z}X^{*} \rightarrow {}^{A}_{Z}X + \text{gamma}(\gamma)\text{rays} \qquad (9.31) $$

Here the asterisk(⋆) indicates the excited state nucleus. In gamma emission, there is no change in the mass number or atomic number of the nucleus.

Boron (${}^{12}_{5}B$) has two beta decay modes as shown in Figure 9.25:

(1) it undergoes beta decay directly into ground state carbon (${}^{12}_{6}C$) by emitting an electron of maximum of energy 13.4 MeV.

(2) it undergoes beta ray emission to an excited state of carbon (${}^{12}_{6}C^{*}$) by emitting an electron of maximum energy 9.0 MeV followed by gamma decay to ground state by emitting a photon of energy 4.4 MeV.

It is represented by

$$ {}^{12}_{5}B \rightarrow {}^{12}_{6}C + e^{-} + \bar{\nu} $$

$$ {}^{12}_{6}C^{*} \rightarrow {}^{12}_{6}C + \gamma $$

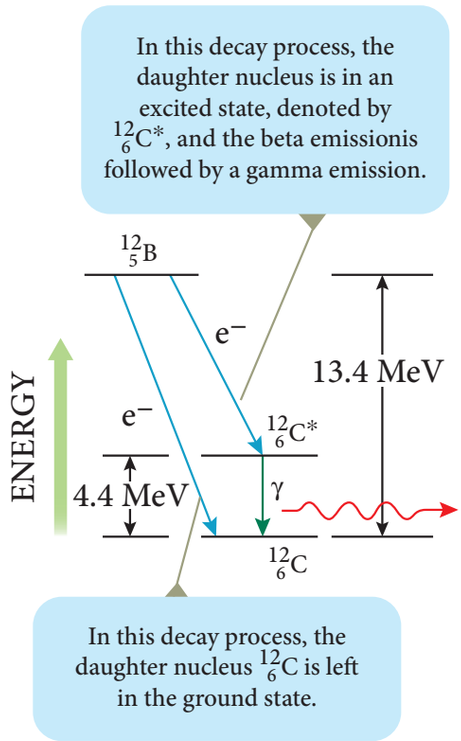


Figure 9.25 Gamma emission

9.6.4 Law of radioactive decay

In the previous section, the decay process of a single radioactive nucleus was discussed. In practice, we have bulk material of radioactive sample which contains a vast number of the radioactive nuclei and not all the radioactive nucleus in a sample

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